

Faunistic study of the sandflies (Diptera: Psychodidae) in an emerging focus of cutaneous leishmaniasis in Al Haouz province, Morocco

S. BOUSSAA^{*,†}, B. PESSON[†] and A. BOUMEZZOUGH^{*}

^{*}Laboratoire d'Ecologie et Environnement, Faculté des Sciences Semlalia, Université Cadi Ayyad, Marrakech, Morocco

[†]Laboratoire de Parasitologie, Faculté de Pharmacie, Université Louis Pasteur-Strasbourg I, B.P. 60024, 74 Route de Rhin, 67401 Illkirch, France

Received 15 July 2008, Revised 15 August 2008,

Accepted 19 August 2008

The Moroccan province of Al Haouz is an emerging focus of human cutaneous leishmaniasis caused by *Leishmania tropica*. In an entomological investigation of three communes in the province (Ourika, Stti Fatma and Rhmate), 3440 sandflies were collected on sticky traps in 2005–2007. Of the six *Phlebotomus* species caught, *Phlebotomus papatasi* (13.3% of the total collection), *P. sergenti* (12.8%) and *P. perniciosus* (12.8%) were predominant, followed by *P. longicuspis* (7.9%), *P. ariasi* (2.6%) and *P. alexandri* (1.1%). All the other sandflies were *Sergentomyia fallax* (24.2%), *S. minuta* (22.0%) or *S. dreyfussi* (3.4%). The distribution of these species among the many biotopes, at varying altitudes (667–1772 m), is discussed.

When the variability of the *P. sergenti* populations in Ourika and Rhmate and those at two sites in northern Morocco (Taounate and Chefchaouene) was investigated by iso-enzyme analysis, seven polymorphic loci (PGI, PGM, HK, MDH1, ME, 6PGD and ACO) were identified. None of the investigated populations of *P. sergenti* was in Hardy–Weinberg equilibrium for all the polymorphic loci. The results indicate that the taxon known as *Phlebotomus sergenti* in Morocco does not have the genetic characteristics of a single species.

In Morocco, three species of *Leishmania* are the causative agents of human leishmaniasis. There is zoonotic cutaneous leishmaniasis caused by *Leishmania major*, which is transmitted by *Phlebotomus* (*Phlebotomus*) *papatasi*, with *Meriones shawi grandis* the main reservoir host (Rioux *et al.*, 1982, 1986a, b). There is also cutaneous and visceral leishmaniasis caused by *L. infantum*, found in humans and domestic dogs and transmitted by *Phlebotomus* species of the subgenus *Larroussius* (Guessous-Idrissi *et al.*, 1997b), and cutaneous leishmaniasis caused by *L. tropica* transmitted by *P. (Paraphlebotomus)*

sergenti (Guilvard *et al.*, 1991). Many Moroccan foci of leishmaniasis have been reported, in, for example, the rural areas of Azilal, Essaouira and Chichaoua (Pratlong *et al.*, 1991; Guernaoui *et al.*, 2005a) and the more urban and peri-urban areas of Taza and Zouagha Moulay Yaacoub (Guessous-Idrissi *et al.*, 1997a; Guessous-Idrissi, 1999; Rhajaoui *et al.*, 2004). Although *L. tropica* has been isolated from a dog in Azilal (Guilvard *et al.*, 1991), the cutaneous leishmaniasis caused by *L. tropica* in Morocco is considered to be anthroponotic, with humans as the only 'reservoirs' (Dereure *et al.*, 1991).

Recently, an emerging focus of cutaneous leishmaniasis in the province of Al Haouz has been described, with the vast majority (90%) of the cases occurring in the com-

Reprint requests to: S. Boussaa, Laboratoire de Parasitologie, Faculté de Pharmacie, Université Louis Pasteur-Strasbourg I, B.P. 60024, 74 Route de Rhin, 67401 Illkirch, France.
E-mail: samiaoussaa@yahoo.fr; fax: +33 3 90244286.

munes of Ourika, Stti Fatma and Rhmate (Anon., 2006; Ramaoui *et al.*, 2008; see Figure). In the present study, the sandflies in these three communes were surveyed over 3 years. The principal objective of this survey was to extend the results and descriptions of Ramaoui *et al.* (2008), by describing the species composition of the sandfly fauna in this new focus, discussing the related ecology, and characterising, by iso-enzyme analysis, four *P. sergenti* populations from Morocco.

MATERIALS AND METHODS

Study Area

The study was conducted at 16 sites in the province of Al Haouz: six in Rhmate (31°25'N, 7°52'W), five in Stti Fatma (31°13'N, 7°42'W) and five in Ourika (31°22'N, 7°51'W). Al Haouz lies at altitudes of 600–1700 m above sea level and has an arid to semi-arid climate, with a mean annual rainfall of 300 mm and monthly mean temperatures that range from 37.7°C in July to 4.9°C in January.

The wild vegetation is composed mainly of jujube (*Zizyphus lotus*) and olive (*Olea europaea*). In the last census, in 2002, the mainly rural province had a total population of 435,090, most of whom were farmers.

Most households in the region have stables nearby and poor sanitation.

Sandfly Collection and Identification

Sandflies were collected in the summers (June–August) of 2005, 2006 and 2007, mainly on sticky traps (20 × 20-cm pieces of paper coated with castor oil). Thirty sticky traps were used at each of the 16 sites. CDC light traps were used at some sites to collect oil-free flies for iso-enzyme analysis and to compare light-trap collections with sticky-trap.

Many biotopes were sampled: intradomestic and peridomestic areas, stables for domestic animals (dogs, chickens and cows), wild areas of vegetation far from human dwellings, caves, drainage holes on roadsides ('barbacanes'), old walls, rock crevices, and ruins.

Each trap was set for one night, from 18.00 hours to 08.00 hours. The sandflies caught on the sticky traps were stored in 70% ethanol and identified to species, by examination of the morphology of the pharyngeal armatures and spermathecae (females) or external genitalia (males), using the keys and descriptions published by the Moroccan Ministry of Health (Anon., 1997). With the members of the *P. perniciosus* complex, the females were identified by examining the

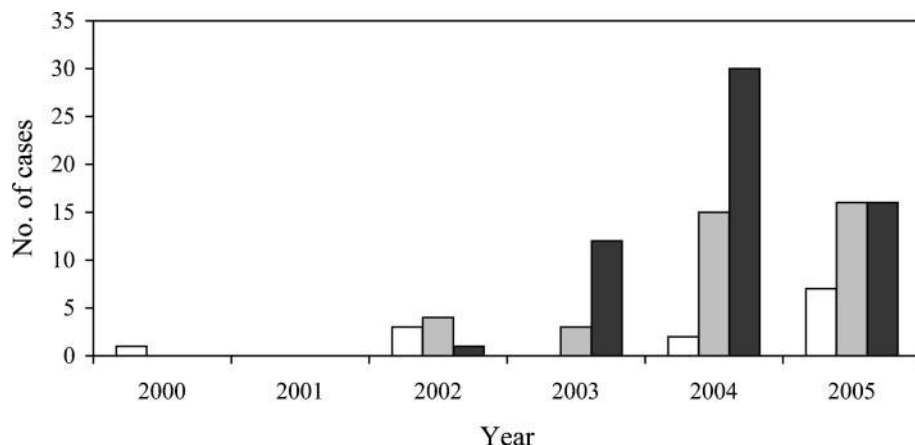


FIG. The numbers of cases of human cutaneous leishmaniasis reported in the Moroccan communes of Rhmate (□), Stti Fatma (■) and Ourika (■) between 2000 and 2005 (Anon., 2006).

dilatation of the distal parts of the spermathecal ducts (Léger *et al.*, 1983) and males by examining the morphology of the copulatory valves (aedeagus) and counting the number of coxite hairs (Benabdennbi *et al.*, 1999; Boussaa *et al.*, 2008).

Iso-enzyme Analysis

The *P. sergenti* caught in light traps in Rhmate (Sidi Bouyahya) and Ourika (Amkhlij) and, for comparison, those caught in similar traps in more northern sites in Morocco — in Taounate (34°36'N, 4°40'W) and Chefchaouene (35°12'N; 5°15'W) — were studied by iso-enzyme analysis.

Iso-electrofocusing was carried out in ultrathin agarose gels, with ampholytes at pH 4.6–5 and pH 3–10 (Boussaa, 2008). The 10 iso-enzyme systems that gave clearly interpretable phenotypes were glucosephosphate isomerase (GPI; E.C.5.3.1.9), phosphoglucumutase (PGM; E.C.5.4.2.2), hexokinase (HK; E.C.2.7.1.1), fumarate hydratase (FUM; E.C.4.2.1.2), malate dehydrogenase (MDH; E.C.1.1.1.37), malic enzyme (ME; E.C.1.1.1.40), 6-phosphogluconate dehydrogenase (6PGD; E.C.1.1.1.44), isocitrate dehydrogenase (ICD; E.C.1.1.1.42), glycerophosphate dehydrogenase (GPDH; E.C.1.1.1.8) and aconitase (ACO; E.C.4.2.1.3).

Data Analysis

Various ecological parameters and indexes were calculated to characterise the sandfly populations in the different sites and habitats. These included 'specific richness' [= S = the number of species in the sample (Spellerberg and Fedor, 2003)] and 'species diversity' [= $(S-1)/\log_e N$, where N is the number of individuals (Sisli, 1997)].

Also calculated was Sorensen's similarity coefficient [= $2a/(2a+b+c)$, where a is the number of species in both sample A and sample B, b the number of species in sample A but not in sample B, and c the number of species in sample B but not in sample A (Krebs, 1989)], which can have values

from 0 (no similarity) to 1.0 (complete similarity).

With the results of the iso-enzyme analysis, allele frequencies were calculated and tested for deviation from those expected of a Hardy–Weinberg (HW) equilibrium, using the BIOSYS-2 program (Swofford and Selander, 1981). As measures of the deviation from HW equilibria and heterozygote excess or deficiency, F_{IS} values were determined (Weir and Cockerham, 1984). In addition, the GENETOP program (Raymond and Rousset 1995) was used to test for genotypic linkage disequilibrium.

RESULTS

Sandfly Collections

During the collections over three summers, a total of 3440 sandflies was collected on sticky traps (Table 1). Nine species — six of *Phlebotomus* (representing 50.47% of the total catch) and three of *Sergentomyia* (49.53%) — were identified in these collections. For each *Phlebotomus* species and for *S. minuta*, the males caught outnumbered the females. Overall, 8.96% of all the female sandflies caught were gravid and another 23.44% were engorged (Table 1).

No obvious morphological anomalies were observed in the sandflies collected but three females collected in Aghbalou were morphologically similar to (but not typical of) *P. ariasi*, and all the male *P. perniciosus* were also considered to be of a slightly atypical form.

Table 2 summarizes the numbers of sandfly species and individuals caught in each category of biotope and the corresponding sandfly biodiversity indices. The sandfly fauna of Sidi Baghdad and Ait Ouaraab showed the greatest biodiversity while that of the centre of Stti Fatma showed the least. Some qualitative and quantitative differences between habitats were noted. Most sandflies of the genus *Phlebotomus* came from intradomestic or peridomestic areas whereas most of the

TABLE 2. Numbers of sandfly species and individuals caught in various habitats in the 16 study sites, and the corresponding biodiversity indices

Code	Site	Elevation	Habitat	No. of sandfly:		Biodiversity index	Dominant species
		(m above sea level)		Species	Individuals		
S1	Sidi Baghdad	811	Intradomestic	6	145	1.02	<i>Phlebotomus papatasi</i>
			Peridomestic	7	132	1.22	<i>P. longicuspis</i>
S2	Sidi Bouyahya	759	Peridomestic	6	239	0.90	<i>P. perniciosus</i>
D1	Dahra	667	Intradomestic	5	85	0.90	<i>P. longicuspis</i>
			Peridomestic	0	0	0	–
A1	Ait ouaarab	764	Intradomestic	4	113	0.64	<i>Sergentomyia fallax</i>
			Peridomestic	7	181	1.15	<i>S. fallax</i>
			Stable	4	60	0.73	<i>S. minuta</i>
A2	Ait belabass	798	Intradomestic	5	165	0.78	<i>S. fallax</i>
			Peridomestic	6	148	1.00	<i>S. fallax</i>
E1	El Hajeb	731	Intradomestic	7	269	1.07	<i>P. longicuspis</i>
S3	Stti Fatma centre	1772	Rock crevices (facing east)	1	34	0	<i>S. minuta</i>
			Rock crevices (facing west)	0	0	0	
			Cave	4	125	0.62	<i>S. minuta</i>
			<i>Barbacane</i>	3	48	0.52	<i>P. ariasi</i>
T1	Tazitount	1247	<i>Barbacane</i>	6	149	1.00	<i>S. minuta</i>
O1	Oulmes	1170	<i>Barbacane</i>	4	77	0.70	<i>P. sergenti</i>
			(facing east)				
			<i>Barbacane</i>	2	23	0.32	<i>S. minuta</i>
			(facing west)				
A3	Aghbalou	1250	<i>Barbacane</i>	5	210	0.75	<i>S. fallax</i>
D2	Tahtdoun	904	Wall	5	139	0.80	<i>S. minuta</i>
L1	Lkhmis	895	Ruin	6	124	1.04	<i>S. minuta</i>
A4	Akhlij	917	<i>Barbacane</i>	6	130	1.02	<i>S. minuta</i>
T2	Tnin Ourika	1000	Vegetation	6	189	0.96	<i>P. papatasi</i>
A5	Akrab	873	<i>Barbacane</i>	4	124	0.62	<i>S. fallax</i>
A6	Amkhlij	846	Wall	6	214	0.94	<i>S. fallax</i>
			Ruin	6	175	0.96	<i>P. perniciosus</i>
			Stable	6	142	1.02	<i>P. papatasi</i>

Sergentomyia collected came from sites further from human habitations, particularly the *barbacanes*. Altitude and aspect within Al Haouz (with the east-facing sites generally having a richer sandfly fauna than the west-facing) may also have affected sandfly species and biodiversity (Table 2).

In terms of Sorensen similarity indices for their sandfly fauna, each of the 16 study sites was fairly similar to the other sites

(Table 3). The lowest levels of similarity observed (with Sorensen similarity indices of <0.4) were those between the centre of Stti Fatma (S3) and Dahra (D1), Akhlij (A4) or Tnin Ourika (T2).

When CDC light traps and sticky papers were used simultaneously at the same sites, the light-trap catches were generally twice as large as the sticky-trap (Table 4). Only two *P. alexandri* were caught in light traps, both in the commune of Ourika.

TABLE 5. Allelic frequencies, with P-values for the deviation from Hardy–Weinberg equilibrium, at each of the polymorphic locus in the four sampled populations of Moroccan *Phlebotomus sergenti*

	Sample from:			
	Taounate	Chefchaouene	Rhmate	Ourika
PGI				
No. of flies	61	30	74	106
Allelic frequency at a pHi of:				
5.43	0.131	0.033	0.000	0.000
5.56	0.016	0.000	0.020	0.000
5.69	0.852	0.967	0.736	0.821
5.85	0.000	0.000	0.027	0.000
5.97	0.000	0.000	0.074	0.000
6.19	0.000	0.000	0.128	0.165
6.51	0.000	0.000	0.014	0.014
P	0.601	1.000	0.245	1.000
PGM				
No. of flies	56	25	34	94
Allelic frequency at a pHi of:				
5.32	0.643	0.880	0.221	0.165
5.41	0.241	0.100	0.662	0.766
5.59	0.063	0.020	0.103	0.064
5.70	0.054	0.000	0.015	0.005
P	0.000	0.000	0.002	0.000
HK				
No. of flies	72	39	78	124
Allelic frequency at a pHi of:				
5.24	0.000	0.000	0.006	0.000
5.28	0.993	0.974	0.968	1.000
5.36	0.000	0.013	0.000	0.000
5.39	0.000	0.013	0.026	0.000
5.49	0.007	0.000	0.000	0.000
P	1.000	0.013	0.001	
MDH1				
No. of flies	33	11	34	66
Allelic frequency at a pHi of:				
5.20	0.939	1.000	0.956	1.000
5.52	0.061	0.000	0.044	0.000
P	1.000		1.000	
ME				
No. of flies	32	7	15	7
Allelic frequency at a pHi of:				
4.55	0.609	1.000	0.967	1.000
4.85	0.391	0.000	0.033	0.000
P	0.000		1.000	
6PGD				
No. of flies	74	37	69	109
Allelic frequency at a pHi of:				
5.85	0.264	0.162	0.167	0.000
5.88	0.682	0.500	0.514	0.679
5.91	0.000	0.014	0.000	0.050
5.94	0.054	0.311	0.319	0.271
6.00	0.000	0.014	0.000	0.000
P	0.000	0.000	0.000	0.000
ACO				
No. of flies	22	27	37	60

TABLE 5. *Continued*

	Sample from:			
	Taounate	Chefchaouene	Rhmate	Ourika
Allelic frequency at a pHi of:				
8.15	0.000	0.019	0.095	0.050
8.30	0.977	0.981	0.892	0.950
8.37	0.023	0.000	0.000	0.000
8.60	0.000	0.000	0.014	0.000
<i>P</i>	1.000	1.000	0.344	0.003

respectively) or for the PGM locus (with F_{IS} values of 0.473, 0.387, 0.384 and 0.814, respectively). The Rhmate and Chefchaouene populations were not in HW equilibrium for the HK locus (with F_{IS} values of 0.794 and 0.490, respectively), the Taounate population was not in such equilibrium for the ME locus ($F_{IS}=0.934$), and the Ourika population was not in such equilibrium for the ACO locus ($F_{IS}=0.649$).

Some significant linkage disequilibria were also detected: between PGM and ME at Taounate ($P=0.00007$), PGM and HK at Chefchaouene ($P=0.04$), PGI and both PGM ($P=0.005$) and HK ($P=0.007$) at Rhmate, and PGM and 6PGD at Ourika ($P=0.001$).

DISCUSSION

Overall, nine species of sandfly were collected and identified in the emerging focus of Al Houaz, not counting the three atypical *P. ariasi*-like females. The latter may have been *P. mariae* since, in Chichaoua, Guernaoui *et al.* (2005b) identified five *P. mariae* males and collected two females that may also have been *P. mariae* from the same biotope.

Species presence and diversity appeared to be affected by several factors. Biodiversity was, for example, relatively low at the highest site investigated (i.e. Stti Fatma, at 1772 m above sea level). In the High-Atlas mountains of Morocco, Guernaoui *et al.* (2006) found the sandfly fauna to be more

diverse at altitudes between 800 and 1199 m than at lower sites (at 400–599 m) or the highest sites they investigated (at 1200–1400 m). In the present study, the east-facing sites seemed to be more diverse, in terms of their sandfly fauna, than the west-facing. It seems unlikely that altitude and aspect have a simple and direct effect on sandfly diversity. Instead, they probably act via their relationship with climate (temperature, precipitation, humidity etc) and geology.

The method used to collect sandflies can also affect the numbers, sex ratios and species of the flies caught, and thus the known distributions of particular species (Lucientes-Curdi *et al.*, 1991). In the present report, although no significant difference was recorded between the CDC light traps and sticky traps in terms of the sex-ratio of the flies caught ($P=0.23$) or the percentage of the females caught that were blood-fed ($P=0.34$), the CDC light traps caught significantly more individuals of each species. In Sanliurfa (Turkey), the male/female and *P. sergenti*/*P. papatasi* ratios for collections made with light traps in individual houses were similar to those for collections made, at the same sites, with sticky papers (Volf *et al.*, 2002). In the same area, 5 years later, light traps were found to collect more individuals of nine sandfly species (of the 13 collected by all methods) than sticky traps, with females making up a larger proportion of the light-trap collections than of the sticky-trap (Toprak and Özer, 2007). In Venezuela, similarly, Wheeler *et al.* (1996) found females making up a significantly larger

proportion of their light-trap collections than of their sticky-trap collections.

In the present study, three *Phlebotomus* species — *P. papatasi*, *P. sergenti* and *P. perniciosus* — made up 39% of the sandflies collected on sticky traps and appeared the most common of the *Phlebotomus* species in the Al Haouz focus. The relative abundance of these three species, all of which can act as vectors of parasites that cause human leishmaniasis, is cause for concern.

Phlebotomus perniciosus is the proven vector of *L. infantum* in the Mediterranean basin (Killick-Kendrick, 1990). Although *L. infantum* usually causes visceral leishmaniasis in humans, not the cutaneous leishmaniasis that is becoming increasingly common in Al Haouz, the Moroccan Ministry of Health has recently been recording one or two cases of human visceral leishmaniasis each year in Al Haouz (Anon., 2006). *Phlebotomus perniciosus* was present and abundant in all three communes investigated in the present study but only in an 'atypical' form (Benabdennbi *et al.*, 1999; Pesson *et al.*, 2004; Boussaa *et al.*, 2008). The epidemiological role of this atypical form, if any, remains to be elucidated.

Although the vector of *L. major* in Morocco is *P. papatasi* (Rioux *et al.*, 1986b) — a species that is not susceptible to infection with *L. tropica* (Killick-Kendrick *et al.*, 1994) — no cases of cutaneous leishmaniasis caused by *L. major* have ever been reported in Al Haouz.

Phlebotomus sergenti is not only highly susceptible to *L. tropica* infection (Killick-Kendrick *et al.*, 1995) but also a proven vector of this parasite (Pratlong *et al.*, 1991). Anthroponotic cutaneous leishmaniasis is restricted to the Old World and largely caused by *L. tropica* transmitted by *P. sergenti* (Desjeux, 2001). Although *P. sergenti* is believed to have a marked 'preference' for semi-arid habitats (Rioux *et al.*, 1984), in the present study it exhibited a wide ecological plasticity, being abundant at 13 of the 16 study sites and in the majority of the nine categories of habitat considered.

It was not found in vegetation distant from households or in rock crevices, probably because the species is endophilic; *P. sergenti* appeared to be the most endophilic of the sandfly species collected in Marrakech city by Boussaa *et al.* (2007).

The taxon known as *P. sergenti* shows considerable genetic variation over its large area of distribution (Depaquit *et al.*, 2002; Yahia *et al.*, 2004; Moin-Vaziri *et al.*, 2007). The possibility that it represents several distinct lineages, possibly including cryptic species, that differ in their vectorial capacities (Depaquit *et al.*, 2002) and/or mechanisms of transmission for leishmanial parasites (Yahia *et al.*, 2004) needs to be considered. In Morocco, three mitochondrial-DNA haplotypes of *P. sergenti* were identified among the sandflies collected in Azilal, Essaouira and Taza, which are *L. tropica* foci (Yahia *et al.*, 2004). These lineages could be markers for (regionally distributed) cryptic species. In a recent study in Spain, two *P. sergenti* lineages were identified — a typically Spanish mitochondrial lineage and another one that is common in Morocco (Barón *et al.*, 2008). In the present study, seven polymorphic loci (PGI, PGM, HK, MDH1, ME, 6PGD and ACO) were identified when *P. sergenti* populations from northern (Taounate and Chefchaouene) and southern (Ourika and Rhmate) Morocco were investigated. The '*P. sergenti*' in these areas does not have the genetic characteristics of a single species, as indicated by the HW disequilibria for PGM, 6PGD and several other loci. The epidemiological significance of this genetic variability merits future investigation.

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